Library Management System in C

Presented By- Tanushree Bali (1262251830)
Pranav Sahoo (1262250194)
Rohit Nukala (1262251929)
Pranav Patil (1262252853)
Mihir Gondse (1262252478)
Antonio Mauro (1262254184)

Objectives

01
Add Books

02
Display Books

03
Search Books

04
Issue Books

05
Return Books

06
Exit System

Arrays Used in the Program

Variable	Purpose
Book_id[10]	Stores IDs of books
Book_name[10][50]	Stores names of books
Book_quantity[10]	Stores quantity available
Count	Tracks number of books

Program Flow (Working)

Main Loop

while(1)

Program runs continuously until user chooses Exit.

Menu Driven Program:

1. Add Books
2. Display Books
3. Search Books
4. Issue Books
5. Return Books
6. Exit

Logic:

- User enters choice.
- Program executes corresponding block
- Returns to menu
- Stops only when choice = 6

Functional Modules

01 Add Book

Takes ID,Name,Quantity
Stores in Arrays
Increments count

02 Display Books

Loops from 0 to count -1
Displays all book details

03 Search Book

User enters ID
Loop checks matching ID
Displays book if found

04 Issue Book

Checks if quantity >0
Decreases Quantity by 1

05 Return Book

Increases quantity by 1

06 Exit

Exits system

DEMONSTRATION

Working Code

```c
#include <stdio.h>
int main()
{
    int book_id[10];
    char book_name[10][50];
    int book_quantity[10];
    int choice;
    int count = 0;


    while(1)
    {
        printf("\n------LIBRARY MANAGEMENT SYSTEM------\n\n");
        printf("1. Add Book\n");
        printf("2. Display Books\n");
        printf("3. Search Books\n");
        printf("4. Issue Book\n");
        printf("5. Return Book\n");
        printf("6. Exit\n\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        // add books
        if (choice == 1)
        {
            printf("\nAdding a Book\n--------------------\n");
            printf("Enter BookID : ");
            scanf("%d",&book_id[count]);

            printf("Enter Book name : ");
            scanf(" %[^\n]",book_name[count]);

            printf("Enter qty: ");
            scanf("%d",&book_quantity[count]);

```

```c
        }

        // display books
        if (choice== 2)
        {
            printf("\n%-5s %-20s %-5s\n","ID","Name","Quantity");
            printf("-------------------------------------------\n");

            for (int i = 0; i < count; i++)
            {
                printf("%-5d %-20s %-5d\n",
                        book_id[i], book_name[i], book_quantity[i]);
            }
        }

        // search book
        if (choice==3)
        {
            int ids;
            printf("\nEnter the ID of the Book: ");
            scanf("%d",&ids);

            for (int i = 0; i < count; i++)
            {
                if (book_id[i]==ids)
                {
                    printf("\nBOOK FOUND! \n");
                    printf("ID: %d\n",book_id[i]);
                    printf("Name: %s\n",book_name[i]);
                    printf("Qty: %d\n",book_quantity[i]);
                }
            }
        }

        // Issue the book
        if (choice==4)
        {
            int id;
            printf("\nEnter the book id you want to issue : ");
            scanf("%d",&id);

            for (int i = 0; i < count; i++)
            {
```

```c
            {
                if (book_id[i]==id)
                {
                    if (book_quantity[i]>0)
                    {
                        book_quantity[i]--;
                        printf("Book issued !!\n");
                        break;
                    }
                }
                else
                {
                    printf("Enter Valid BookID!\n");
                }

            }
        }

        // Return book
        if (choice==5)
        {
            int id;
            printf("\nEnter the Book id : ");
            scanf("%d",&id);

            for (int i = 0; i < count; i++)
            {
                if (book_id[i]==id)
                {
                    book_quantity[i]++;
                    printf("Book Returned !!\n");
                }
            }
        }

        // Exit
        if(choice == 6)
        {
            break;
        }
    }
```

OUTPUT

ADDITION OF BOOK

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display Books
3. Search Books
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 1

Add a book to the shelf:-
Enter book id :- 1234
Enter book name :- Rohit
Enter quantity:- 4

Book added successfully!!

Number of Rohit  books available now are :- 4
```

DISPLAY BOOKS

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display Books
3. Search Books
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 2

ID    Name                Quantity
---------------------------------------------
1234  Rohit               4
```

SEARCH BOOK

```
Enter the ID of the Book: 1234


BOOK FOUND!
ID: 1234
Name: Rohit
Qty: 4
```

OUTPUT

ISSUE BOOK

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display Books
3. Search Books
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 4

Enter the book id you want to issue : 1234
Book issued !!
```

RETURN BOOK

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display Books
3. Search Books
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 5

Enter the Book id : 1234
Book Returned !!
```

EXIT

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display Books
3. Search Books
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 6


=== Code Execution Successful ===
```

Thank You